

Vedant Modi

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EDUCATION

Tufts University

August 20

GPA 3.74, BS Computer Science, BS Mathematics, Magna Cum Laude, Dean's List (Fall 2022, Spring 2024, Spring 2025, Fall 2025)

Relevant Coursework:

Distributed systems, Computer architecture, Algorithms, Embedded systems, Machine learning, Data structures, Statistics/Probability theory, Operating systems, Compilers, Mathematical modeling, Abstract algebra, Programming language theory, Network security, Linguistics

SKILLS

Programming Languages: C, C++, Rust, Python, Java, TypeScript/JavaScript, Go, Embedded C, x86-64 Assembly, OCaml, Standard ML, Scala
Systems & Infrastructure: Operating Systems, Distributed Systems, Event-driven Systems, Network Programming, Concurrency, Performance Optimization, Compiler Engineering, Docker, CI/CD, Git (version control, peer code review)
Databases & Web: PostgreSQL, MongoDB, Node.js, Spring Boot, React, HTML, Tailwind CSS, Figma
Testing & Engineering Practices: Unit Testing, Integration Testing, Regression Testing, Test Automation
Certifications: AWS Certified Cloud Practitioner
Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib

RELEVANT EXPERIENCE

Software Engineering Intern – Markets Technology, Commercial & Investment Banking

June 2025 – August 2025

J.P. Morgan Chase & Co.

New York, NY

- Enabled performance analytics for over **1,000,000+ transactions per day** across **5 critical data enrichment services** by building and deploying a production monitoring application for high touch bond trading platform.
- Flagged **20% of requests** suffering extreme latency in key service, driving targeted optimizations that significantly improved trade execution workflow in global markets.
- Prototyped an AI-powered travel recommendation tool later adopted by firm, leveraging an agentic RAG workflow to give detailed user suggestions and promotional feedback. Recognized among **the top 5 teams** in New York City offices for effective presentation and innovative product.
- Expanded scope of product ownership by iterating on features and incorporating feedback from stakeholders in **5+ widespread locations**. Enhanced the application to provide actionable metrics for **99% of requests**, iterated on live troubleshooting features, and ultimately achieved **100% alignment with the product vision**.
- Brought application from concept to completion **within 10 weeks** using the firm SDLC; managing feature development, testing, stakeholder approval, and critical business periods such as code freezes. Enabled future work within the SDLC by creating a maintainable and well-documented codebase.

Teaching Fellow for Machine Structure & Assembly Programming

January 2025 – May 2026

Tufts University, Department of Computer Science

Somerville, MA

- Mentored **200+ students'** ability to engineer large-scale, low-level programs by encouraging rigorous testing, building modular architecture, creating powerful data abstractions, writing strong documentation, and harnessing standard libraries.
- Enriched **200+ students'** developer soft skills (i.e. pair programming, product ownership) by introducing one-on-one code reviews.

Teaching Assistant for Data Structures, Machine Structure & Assembly Programming

May 2023 – December 2024

Tufts University, Department of Computer Science

Somerville, MA

- Strengthened course infrastructure for **200+ students** by improving autograding software, staff software, and assignment solutions; contributed **20+ unit tests** to autograder by finding edge behavior in students' submissions.
- Graded assignments by studying **200+ submissions** for functionality, testing, and course coding standards.
- Reinforced course objectives and debugging principles during **2500+ student interactions** by explaining lecture topics and assignments in office hours.

Full Stack Developer

September 2023 – May 2024

Tufts JumboCode

Somerville, MA

- Enhanced the information display for the **1,000,000+ annual visitors** of the Emerald Necklace Conservancy by designing a full-stack iOS/web application in a tight-knit, agile team
- Created a secure page modification system for park administration by maintaining a MongoDB database for information, and an authentication system for editing privileges
- Automated database maintenance using PyMongo, reducing manual data handling tasks, and speeding up database work **by 30%**
- Designed a cohesive user interface for **50+ pages** by creating and documenting React components in TypeScript

Receptionist

May 2023 – Present

Tufts University, Office of Academic Space Management

Somerville, MA

- Service community via organizing mail, and analyzing building usage for largest academic buildings on Tufts campus.

RELEVANT PROJECTS

Photos by Vedant Modi | Next.js, TypeScript, Rust, Cloudflare Workers, PostgreSQL

August 2025 – Present

- Built a full-stack photo journal plotting **400+** geotagged photos backed by a serverless **PostgreSQL** database and a global content delivery network.
- Improved data throughput by building an ingestion pipeline that concurrently processes EXIF metadata and multi-resolution image compression, leveraging Rust's performance and memory safety guarantees.
- Reduced ingest time for users by eliminating manual caption writing, deploying a vision model that generates a caption for each image upon ingestion and passing the result to users for approval.

Real-Time Transit Data Pipeline | Go, Embedded C

March 2026 – April 2026

- Delivered real-time NYC subway arrivals across all 24 lines to an OLED display by building a serverless Go function that deserializes a live protobuf feed and serves parsed JSON to an ESP32, which forwards arrival data to an STM32 over UART for rendering.

- Processed streaming transit data under embedded memory and latency constraints, marshaling between cloud, Wi-Fi, and UART boundaries prior to device rendering.

Optimizing Functional Language Compiler | *Scala, x86-64 Assembly, Java*

January 2026 – April 2026

- Built an optimizing compiler for a functional language transforming source programs through a multi-stage pipeline down to machine-independent assembly by applying classical compiler optimizations over a CPS intermediate representation.

Universal Machine | *C, x86-64 Assembly, Bash*

November 2023 – December 2023

- Created a Turing Complete virtual machine by **implementing a custom memory subsystem** with a segment table supporting dynamic allocation and deallocation, separated functionality using **object-oriented programming techniques** into I/O, machine arithmetic, and logic; validated correctness with a custom unit-testing framework.
- Optimized the program by analyzing x86-64 Assembly instructions and **qcachegrind** and minimized expensive operations such as dereferencing or allocation through reuse of memory; verified performance gains via benchmarking against **1,000,000,000+ instruction** binaries

Movie Recommendation System | *Python, Scikit-learn*

April 2025

- Generated personalized movie recommendations for 100K+ user-item interactions by implementing collaborative filtering and latent factor models on the MovieLens100k dataset, evaluated using RMSE and ranking-based metrics across matrix factorization configurations.

File Copy | *C/C++, Bash*

September 2025 – October 2025

- Developed a fault-tolerant network file transfer application, as measured by error-free transfer on unreliable systems, by designing a packet-level protocol over UDP that handles retransmissions, timeouts, and disk-write errors.
- Ensured protocol reliability by stress-testing 10+ GB directory transfers under packet duplication, drops, reordering, and artificial delay, as well as simulated disk-level faults like slow or failed block reads/writes.

RPC Generator | *C/C++, Python, Bash*

November 2025

- Built a transparent remote procedure call framework supporting arbitrary C++ function prototypes, verified by successful cross-network invocation of functions with complex argument types including nested structs and arrays.
- Ensured reliable structured data recovery across TCP's stream abstraction by designing a text protocol with explicit length-prefixed fields and partial-read handling — mirroring serialization challenges in production distributed systems.

How HTTP/3 and QUIC Overcome TCP's Leaky Abstraction | *Technical Writing, Networking protocols*

December 2025

- Analyzed the evolution of the HTTP protocol by researching TCP head-of-line blocking, QUIC's role in improving mobile web performance in HTTP/3, the security implications of mandatory encrypted communication, and the backward-compatibility decisions that enabled rapid adoption at scale.

EXTRACURRICULAR ACTIVITIES

Spoken Languages: Proficiency in English, Hindi, Urdu, Spanish, and French

Media: Droneography, Photoshop, Lightroom, After Effects, Davinci Resolve Studio, Premiere Pro, InDesign, Wordpress